

Homework 4

Please read Section 1.1.1 of Durrett, through the end of Example 1.3. This section is technical and includes the extra rule of probability (having to do with situations where there are infinitely many outcomes). Example 1.3 is a situation where it is natural not to consider all outcomes to be equally likely.

1. In Homework 3, we studied the chance experiment of rolling a red die and a green die. To understand the experiment, we used the sample space Ω (with 36 equally likely outcomes) from Example 1.2 in Durrett. Let A be the event “the number on the red die is larger than the number on the green die,” and let B be the event “the numbers on the two dice differ by at most 1.”
 - a. List the outcomes in the event $A \cup B$. That event is the “union” of A and B , the set of all outcomes that are in A or in B (or both). In real-world terms, these are the outcomes where the number on the red die is larger than the number on the green die or the numbers on the two dice differ by at most 1 (or both).

We can use the sample space from Durrett's Example 1.2. The outcomes in that sample space are ordered pairs like $(1, 2)$ or $(5, 4)$. We interpret the first number as the roll on the red die and the second number as the roll on the green die.

$A \cup B = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), (5, 1), (5, 2), (5, 3), (5, 4), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (1, 2), (2, 1), (2, 3), (3, 2), (3, 4), (4, 3), (4, 5), (5, 4), (5, 6), (6, 5)\}$

I typed first the elements of A and next the elements of B . I then crossed out the five repeated elements. That left 26 elements.

- b. List the outcomes in the event $A \cap B$. That event is the “intersection” of A and B , the set of all outcomes that are both in A and in B . In real-world terms, these are the outcomes where the number on the red die is larger than the number on the green die and the numbers on the two dice differ by at most 1.

$A \cap B = \{(2, 1), (3, 2), (4, 3), (5, 4), (6, 5)\}$

The outcomes in the intersection $A \cap B$ are exactly the five repeated elements that I crossed out in 1a.

- c. Are A and B disjoint events?

No: “disjoint” would mean $A \cap B = \emptyset$, i.e. that there are no elements in both A and B . As we saw in 1b, these events have five outcomes in common.

- d. List the outcomes in the event A^c . That event is the “complement” of A, the set of all outcomes that are not in A. In real world terms, these are the outcomes where the number on the red die is not larger than the number on the green die.

$$A^c = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 3), (3, 4), (3, 5), (3, 6), (4, 4), (4, 5), (4, 6), (5, 5), (5, 6), (6, 6)\}$$

There are 21 outcomes in A^c .

- e. Calculate $P(A \cap B)$, $P(A) \cdot P(B)$, $P(A \cup B)$, $P(A) + P(B)$, $P(A)$, and $P(A^c)$.

$$P(A) = 15/36, P(B) = 16/36$$

$$P(A \cap B) = 5/36$$

$$P(A) \cdot P(B) = 240 / 1296 = 5 / 27.$$

$$P(A \cup B) = 26/36 = 13/18$$

$$P(A) + P(B) = 31/36$$

$$P(A^c) = 21/36$$

2. On the first day of class, we thought about the New York State LOTTO game. In that game, you pick six numbers from 1 – 59. The state lottery commission draws six numbers at random from the same list. If your six numbers match their six, you win the jackpot. The order of the numbers picked does not matter, just the set of six numbers.

We will think of this lottery as a chance experiment. The random event is drawing the six numbers. The sample space is all the possible sets of six numbers chosen from 1 – 59. All outcomes in this sample space are equally likely. We counted these outcomes and found $C_{59,6}$ of them.

- a. Suppose your numbers are 3, 14, 15, 9, 26, 53. What is the probability that you win the jackpot? In other words, what the probability of the event that has just one outcome, your numbers {3, 14, 15, 9, 26, 53}?

The sample space for the random draw has $C_{59,6}$ outcomes, and all are equally likely. The probability of winning the jackpot is $1/C_{59,6} = 1/45,057,474 \approx 0.0000000222$.

- b. Consider the event “the random draw includes exactly three of your numbers.” How many outcomes are in this event? (Multiplication rule: first pick three of your numbers that will be in the random draw, then ...)

We can count draw outcomes that match exactly three of your numbers using the multiplication rule: to make such an outcome, first choose three of your six numbers that will match the drawn numbers ($C_{6,3}$ possibilities). There are 53 (= 59 – 6) numbers that are not on your list

(everything except 3, 14, 15, 9, 26, 53). To complete the outcome, choose three of these 53 numbers ($C_{53,3}$ possibilities). Combining these steps with the multiplication rule, we find $C_{6,3} C_{53,3} = 468,520$ outcomes that match exactly three of your numbers.

- c. What is the probability that the random draw includes exactly three of your numbers? You can look this probability up on the lottery web site. The site lists it as “1 in 96.17” or about 0.0103983. If your probability value is not very close to that number, check your work in part b.

Since each of the $C_{59,6}$ outcomes of the draw is equally likely, the probability is $C_{6,3} C_{53,3} / C_{59,6} \approx 0.0103983$.

3. Imagine your roll a fair six-sided die six times. This is your chance experiment.

- a. What is a good sample space Ω to use for thinking about this experiment? How many outcomes are there in your sample space? (This situation is like the die-rolling activity from August 26th, which we discussed on August 28th.)

A good sample space Ω is the set of all sequences of six whole numbers between 1 and 6. The outcome written as (3, 3, 1, 6, 5, 4), for example, means that your first roll was a 3, your second was a 3, your third was a 1, and so forth. By the multiplication rule, there are $(6)(6)(6)(6)(6)(6) = 6^6 = 46,656$ outcomes in the sample space. We would never try to write all these outcomes!

- b. What is the probability that you roll all six numbers on your six rolls? Give your value as an exact fraction and as a decimal approximation, rounded to the nearest thousandth (like 0.123).

An outcome where we roll all six numbers is a permutation of the six numbers 1, 2, 3, 4, 5, 6 such as (3, 1, 2, 4, 6, 5) or (6, 5, 4, 3, 2, 1). There are $6!$ such permutations. Therefore, the probability is $6! / 6^6 \approx 0.015$.

- c. What is the probability that you repeat at least one number on your six rolls? Give your value as an exact fraction and as a decimal approximation, rounded to the nearest thousandth.

We can use the familiar “at least” technique. There are $6^6 - 6! = 45,936$ with at least one repeated number. The probability of getting a repeated number is thus $(6^6 - 6!) / 6^6 = 45936 / 46656 \approx 0.985$.

Let A be the event “we roll all six numbers.” Then A^c is the event “we repeat at least one number in our six rolls.” A shortcut to calculate $P(A^c)$ is to use the formula $P(A^c) = 1 - P(A)$. That formula gives the values mentioned in the previous paragraph.

4. Please work Exercise 6 from Chapter 1 of Durrett.

If we flip a coin 5 times, what is the probability that the number of heads is an even number (that is, divisible by 2)?

We can calculate this probability using a suitable sample space. Outcomes for this chance experiment (flipping a coin 5 times) are sequences of five H's (heads) and T's (tails) such as HTHHT or TTTTH. All such outcomes are equally likely.

Let A be the event "the number of heads is an even number." This event is the disjoint union of three events: "flip zero heads," "flip two heads," and "flip four heads."

There is 1 outcome in the event "flip zero heads" (namely, TTTTT). We can think of that "1" as $C_{5,0}$, counting the ways to choose 0 slots for H's out of the 5 flips.

There are $C_{5,2} = 10$ outcomes in the event "flip two heads": we count the ways to choose 2 slots for H's out of the 5 flips.

There are $C_{5,4} = 5$ outcomes in the event "flip four heads": we count the ways to choose 4 slots for H's out of the 5 flips.

In total, there are $1 + 10 + 5 = 16$ outcomes in the event A , and so $P(A) = 16 / 32$ or $1 / 2$, since all the outcomes in Ω are equally likely.